Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/26/2006
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Crop and Bio Mass Production		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cmitchel@purdue.edu	**Fax:**	FY 765-494-0391
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-1347
Organization Name:	Purdue University		
PI Address 1:	Department of Horticulture and Landscape Architecture		
PI Address 2:	625 Agriculture Mall Dr		
PI Web Page:	http://www.alsnscort.org		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Beyl, Caula (Alabama A & M) Mitchell, Cary (Purdue University) Brown, Paul (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	Three NSCORT (NASA Specialized Center of Research and Training) projects address the feasibility of producing edible biomass in a space regenerative life-support system. One project addresses a novel approach to significantly reduce the ESM (Equivalent System Mass) of growing food crops in controlled environments; the second project tests the potential to use fungi to degrade non-edible crop residues while generating edible mushrooms; and a third project addresses the feasibility of feeding crop or bioreactor residue to fish that can be eaten. A novel lighting system for higher plants has demonstrated incremental improvement in electrical energy utilization during the project by correcting software glitches, by creating growth compartments that reflect escaped photons back into the crop stand, by planting dense crop stands that close fast, by positioning plants directly in front of one-sided light strips, and by relying upon the “plasticity” of plant growth to seek out light. Appropriate red/blue ratios to maintain this “light magnet” are being		

Task Description:	defined. Intracanopy lighting is superior to overhead lighting for crop quantum efficiency and electrical-energy utilization once foliar canopies have closed. It has been confirmed that reproductive development occurs normally with intracanopy LED lighting. Remote position sensing of canopy height and automated switching of light engines is in development. Once needed engineering upgrades to the prototype intracanopy lighting system are implemented, ESM for crop lighting should be reduced at least an order of magnitude relative to that of traditional overhead lighting systems. The fungi/mushroom project has focused on optimizing conditions for using crop residues as growth substrates. Fungi have a reputation for enzymatic degradation of polymers that are recalcitrant to bacteria. However, the fungal species tested attack the most easily digestible substrates first, before taking on more recalcitrant ones. This finding has led to thinking that microbial bioreactors might first be used to remove soluble metabolites and more readily degradable plant polymers from crop residue, such as hemi-cellulose, pectin, and cellulose that is not sequestered by lignin. Fungal composters or bioreactors, then, will be challenged from the start with polymers that require induction of enzymes that are not otherwise synthesized when the fungi are not so challenged. After first-stage metabolic by-products (CH_4, CO_2, NH_3) are removed, fungal hyphae and fruiting bodies would be forced to grow on recalcitrant residues. Because crop fibrous substrates are nitrogen and sulfur-deficient, an evolving strategy is to loop back appropriate amounts of completely degraded, gaseous effluents from non-plant bioreactors or physico-chemical waste treatments. Unfinished business from the fungal-composting project is to match the right kinds of crop residues as substrates with the right fungal species under appropriate culture conditions. Potential for fungi to play a dual role in a regenerative life-support system by lowering the ESM of crop-waste degradation while creating alternative food and feed sources remains high. Fresh-water Tilapia have had a reputation for being able to digest cellulose, which humans cannot do. That was the initial rationale for considering fish both for crop-waste reduction and as an alternative food source for space crews. Although the initial hypothesis proved to be a myth, Tilapia are readily able to use microbial biomass as a food source, and those bacteria have more capability for degrading plant polymers in a bioreactor than the fish do. Thus, a role for Tilapia as an alternative food source becomes a food-chain issue, and just as for fungi, looping back completely degraded N- and S-rich gaseous effluents from processing of non-plant wastes to bacterial crop-waste reactors provides essential substrates for protein and fat synthesis in fish biomass. Closing the loop between fish, fungi, and bioprocessing of crop and other wastes remains to be fully integrated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Development of solid-state, intracanopy and close-canopy crop-lighting systems will solve the light-distribution problem that has long limited the photosynthetic productivity of overhead-lighted crop stands, and will permit the use of low-power light sources. These advances will enable year-round, warehouse-based, controlled-environment agriculture (CEA) production of salad greens, herbs, and certain vegetables to be profitable in local markets, especially if coupled with sources of cheap, bio-derived energy, and competitive with produced trucked from distant production sites. 2. Further development of biological methods for converting organic wastes into fuels, chemicals, and other useful products. 3. Developing methods for maximum crop yield at minimum cost in controlled-environment agriculture (CEA) 4. Developing genetic engineering of CEA crops for superior performance
	BioMass Production: LED Lighting Project: Second prototype array. Since our initial NSCORT research with a first prototype LED lighting array, we have developed and obtained a second array. Significant upgrades were instituted in the second array. First, half of the individual lighting strips or 'lightsicles' were constructed with an additional 17.8 cm of length in the tubular support located between the top light engine and the bottom of the electronics enclosure on each lightsicle. This allows the shorter lightsicles to nestle beneath the electronics enclosure of the longer lightsicles in the intracanopy (IC) configuration, thus allowing closer grouping of lightsicles and a much wider variety of possible configurations. Reconfiguring the IC arrangement led to more uniform distribution of light within the canopy. A second significant upgrade to the LED array was a software modification developed by Orbitec. This modification caused a change in the allocation of power to particular drivers, which resulted in the ability to obtain uniform light levels from a particular light engine regardless of the number of other engines energized on the same strip. LED Lighting Project: Side-by-side intracanopy versus overhead experiments. Following five trials with the first LED array in either the IC or OH configuration, the original array has been maintained in the OH configuration, and the second prototype is in the IC configuration. The first of these experiments was conducted using cowpea, with two groups of 30 plants (129 plants/m2) grown simultaneously for 32 days. The data clearly demonstrate that plants grown under the OH array were less productive than plants grown with IC lighting. Even more impressive are the numbers for grams dry weight per Kilowatt-hour of electricity consumed, 1.14 for the IC plants compared to 0.67 for the OH-lit plants. This value for the IC-grown plants also exceeds the previous maximum obtained in Trial 4 (0.98 g/kW-h), although the total g DW decreased (from 96.37 g in trial 4 to 84.46 g in the IC comparison). This, coupled with a decrease in the average g DW per plant, indicates that competition within the more dense canopy for light and other resources was occurring. Also, during this trial it was observed that the plants grown with the IC lighting displayed some degree of leaf intumescence growth, while those grown under the OH array did not. The higher percentage of blue light in the OH array may have inhibited the formation of intumescence growths on the leaves. Since intumescence had never previously been observed on the plants grown in trials 1-5, it is likely that the blue:red ratio was high enough for those trials as well, although this ratio likely changed with increasing numbers of light engines energized. Based on the findings of these experiments, thorough analysis of blue and red levels, independently of each other, were made for each array stepwise as each light engine was energized. In the interest of determining the potential of LEDs for crop production involving reproductive development, a side-by-side IC vs. OH comparison using 'Triton' pepper was begun on April 28, 2006. After almost one month of growth, the plants showed severe intumescence symptoms, and at that time the LED settings were adjusted to provide 15% of the incident PAR from the blue LEDs. This adjustment did not appear to mitigate the intumescence symptoms. Harvesting continues and initial results show that flowers and fruit form and ripen normally under both lighting conditions, in spite of intumescence.

Task Progress:	LED Lighting Project: Gas exchange. The design of the Minitron 3 gas-exchange cuvette system for intracanopy LED lighting of small crop stands is nearly complete. A transpiration study was conducted in conjunction with the first side-by-side IC versus OH experiment discussed earlier. This experiment culminated in an ICES paper and will be discussed in a report in the systems group focus area. Extending Crop Harvest Index Using Edible Fungi Experiments were conducted to determine best wheat straw particle size (coarse, medium, fine) for closed system fungal culture, assess various sources of nitrogen for enhancing growth in select strains of shiitake (Lentinula edodes) and maitake (Grifola frondosa), and to investigate cropping patterns for greater mycelial colonization and degradation of various crop residues. Fine textured wheat straw (particles pass through 2mm sieve) was most promotive to fungal growth. Our studies demonstrated conclusively that the oyster mushroom (Pleurotus ostreatus) strain 'Grey Dove' is best for degrading lignocellulosic wastes and recycling food residues and sludge. A 72% or higher degradation efficiency was achieved by recomposting spent mycelia and residual undergraded crop material with strain 'Grey Dove'. Tilapia Fish In Space. Utilization of wastes. Working with collaborators associated with the Center, waste products were collected. Wastes evaluated included a wheat bran/wheat germ mixture, non-edible biomass from soybean (SB), sweet potato (SP), cowpea (CP) and basil (B), and bacterial residue from an autothermal thermophilic aerobic digester (TB). Experimental diets consisted of 98% waste product and 2% binder to facilitate pelleting. All diets were accepted by tilapia and all treatment groups gained weight. Our goal was to continue characterizing waste products as colleagues evaluated various plant and food sources, then develop a balanced diet for tilapia using the chemical characterization data and test our mixed diets in a subsequent study in year 5. Growth was inversely related to fiber concentrations and directly related to methionine concentration, an essential amino acid. Composted waste. Cowpea waste was acquired from colleagues at Purdue and shipped to Alabama A&M for composting. Based on these data, composting with the oyster mushroom did not significantly alter the fiber component of the residue. Feeding either product resulted in negative weight gain in juvenile tilapia First-feeding diets. One of the more important considerations, and one that had not been addressed in research studies with tilapia, was their ability to use plant-based diets at first feeding. We developed an all-plant ingredient basal diet for tilapia and that diet appeared deficient in three essential amino acids (methionine, threonine and lysine). Based on these data, first feeding tilapia can be fed plant-based diets without sacrificing weight gain. Mineral balance in aquaponics. Based on observations from an established aquaponic system, boron, iron and zinc appeared deficient for plant growth in integrated aquaponic systems. However, plant growth was suboptimal in this initial study. This subcomponent was used as a training module. Data developed are preliminary in nature, and were going to be used for a more formal evaluation.

Bibliography Type:	Description: (Last Updated: 04/16/2020)
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